

Appendix B

The realized beta estimator in (3) implicitly assumes that, for each day t , we record all asset prices and factors at the same frequency j/M , with $j = 1, \dots, M$. In practice, however, we observe asset prices and factors only up to a market microstructure noise, and also at time intervals of irregular lengths. The assumptions of synchronicity and absence of microstructure noise are perhaps reasonable for a low enough intraday frequency (say, 30 minutes), but not at the ultra-high frequency. To account for both nonsynchronous trading and microstructure noise, we use Barndorff-Nielsen et al.'s (2011) multivariate realised kernel approach in the empirical sections. In what follows, we briefly describe the realized kernel estimator for the factor loadings and then establish the conditions under which Propositions 1 to 3 remain valid.

We start with the notion of refresh time (Harris, McNish, Shoesmith and Wood, 1995; Barndorff-Nielsen et al., 2011). For each asset i , the first refresh time $\tau_{i,t,1} \in [t, t+1]$ is the time it takes to observe the opening trades of the asset and of all factors. We now define $\tau_{i,t,2} \in (\tau_{i,t,1}, t+1]$ as the minimum time it takes to observe at least one additional trade for the asset i as well as for all factors. We proceed computing subsequent refresh times $\tau_{i,t,\ell+1} \in (\tau_{i,t,\ell}, t+1]$ with $\ell = 2, 3, \dots$ until we reach the end of day t . This results in a set of refresh times $(\tau_{i,t,1}, \tau_{i,t,2}, \dots, \tau_{i,t,M_{i,t}})$, so that $M_{i,t}$ corresponds to the effective sample of intraday observations that we have to estimate the daily factor loadings. The crucial assumption is that the refresh times are independent of asset prices and returns.

Further, to allow for market microstructure noise, we consider $\tilde{P}_{i,\tau_{i,t,\ell}} = P_{i,\tau_{i,t,\ell}} + \eta_{i,\tau_{i,t,\ell}}$ and $\tilde{F}_{j,\tau_{i,t,\ell}} = F_{j,\tau_{i,t,\ell}} + \eta_{F_j,\tau_{i,t,\ell}}$ for $\ell = 1, \dots, M_{i,t}$ and $j = 1, \dots, k_F$. The error terms $\eta_{i,\tau_{i,t,\ell}}$ and $\eta_{F_j,\tau_{i,t,\ell}}$ may correlate respectively with $P_{i,\tau_{i,t,\ell}}$ and $F_{j,\tau_{i,t,\ell}}$ as well as exhibit some autocorrelation as in ? Lemma 1(iii) or as in Barndorff-Nielsen et al. (2011) Assumption U. Although we further allow for correlation between $\eta_{i,\tau_{i,t,\ell}}$ and $\eta_{F_j,\tau_{i,t,\ell}}$, the orthogonality of the factors rule out any correlation between the factor noises, that is to say, $\eta_{F_j,\tau_{i,t,\ell}}$ and $\eta_{F_k,\tau_{i,t,\ell}}$ are orthogonal for $1 \leq j \neq k \leq k_F$.

We now define the realized kernel estimator of the j -th factor loading for asset i at dat t as

$$\hat{\beta}_{i,t,M_{i,t}}^{(j)} = \frac{\sum_{h=1}^H \kappa\left(\frac{h-1}{h}\right) \left(\gamma_{t,h}^{\tilde{P}_i, \tilde{F}_j} + \gamma_{t,-h}^{\tilde{P}_i, \tilde{F}_j} \right)}{\sum_{h=1}^H \kappa\left(\frac{h-1}{h}\right) \left(\gamma_{t,h}^{\tilde{F}_j} + \gamma_{t,-h}^{\tilde{F}_j} \right)}, \quad \text{for } j = 1, \dots, k_F$$

where κ is a twice continuously differentiable kernel function such that $\kappa(0) = 0$ and $\kappa'(0) = 0$,⁶

$$\begin{aligned}\gamma_{t,h}^{\tilde{P}_i, \tilde{F}_j} &= \sum_{\ell=H}^{M_{i,t}-H-1} \left(\tilde{P}_{i, \tau_{i,t, \ell+1}} - \tilde{P}_{i, \tau_{i,t, \ell}} \right) \left(\tilde{F}_{j, \tau_{i,t, \ell+1-h}} - \tilde{F}_{j, \tau_{i,t, \ell-h}} \right) \\ \gamma_{t,h}^{\tilde{F}_j} &= \sum_{\ell=H}^{M_{i,t}-H-1} \left(\tilde{F}_{j, \tau_{i,t, \ell+1}} - \tilde{F}_{j, \tau_{i,t, \ell}} \right) \left(\tilde{F}_{j, \tau_{i,t, \ell+1-h}} - \tilde{F}_{j, \tau_{i,t, \ell-h}} \right).\end{aligned}$$

Assume now that $\Sigma_{FF,t}^{(j,j)}$, $\Sigma_{i,t}^{(j)}$, and $\Sigma_{F,t}^{(j)}$ are bounded uniformly in t for $j = 1, \dots, k_F$. It then follows from Theorem 3 and Corollary 1 in Barndorff-Nielsen et al. (2011) that

$$\begin{aligned}\hat{\beta}_{i,t, M_{i,t}}^{(j)} - \beta_{i,t}^{(j)} &= \frac{1}{\Sigma_{FF,t}^{(j,j)}} \left[\sum_{h=1}^H \kappa \left(\frac{h-1}{h} \right) \left(\gamma_{t,h}^{\tilde{P}_i, \tilde{F}_j} + \gamma_{t,-h}^{\tilde{P}_i, \tilde{F}_j} \right) - \Sigma_{i,t}^{(j)} \Sigma_{F,t}^{(j)} \right] \\ &\quad + \frac{\sum_{h=1}^H \kappa \left(\frac{h-1}{h} \right) \left(\gamma_{t,h}^{\tilde{F}_j} + \gamma_{t,-h}^{\tilde{F}_j} \right) - \Sigma_{FF,t}^{(j,j)}}{\sum_{h=1}^H \kappa \left(\frac{h-1}{h} \right) \left(\gamma_{t,h}^{\tilde{F}_j} + \gamma_{t,-h}^{\tilde{F}_j} \right) \Sigma_{FF,t}^{(j,j)}} \sum_{h=1}^H \kappa \left(\frac{h-1}{h} \right) \left(\gamma_{t,h}^{\tilde{P}_i, \tilde{F}_j} + \gamma_{t,-h}^{\tilde{P}_i, \tilde{F}_j} \right),\end{aligned}$$

for $H = H_{i,t} \simeq cM_{i,t}^{3/5}$ and some positive constant c . This implies that $\mathbb{E} \left(\hat{\beta}_{i,t, M_{i,t}}^{(j)} - \beta_{i,t}^{(j)} \right) = O_p \left(M_{i,t}^{-1/5} \right)$ and that $\text{Var} \left(\hat{\beta}_{i,t, M_{i,t}}^{(j)} - \beta_{i,t}^{(j)} \right) = O_p \left(M_{i,t}^{-2/5} \right)$. Alternatively, we could set $H = H_{i,t} \simeq cN_{i,t}^{1/2}$ for some positive constant c and choose a kernel such that $\kappa''(0) = 0$. This would entail, by ? Lemma 1, an average realized beta estimation error of order $O_p \left(M_{i,t}^{-1/4} \right)$, with a variance of order $O_p \left(M_{i,t}^{-1/2} \right)$. However, the drawback is that the realized kernel estimator of the covariance matrix with $\kappa''(0) = 0$ is not necessarily positive definite.

Now, under the additional assumption that $\mu_{i,t}$ and $\mu_{F_j,t}$ are bounded uniformly in t , the contribution of the drift component to $\frac{1}{Th^k} \sum_{t=1}^{T-1} f_{j,t+1} \left(\hat{\beta}_{i,t, M_{i,t}}^{(j)} - \beta_{i,t}^{(j)} \right)$ is of order $O \left([\inf_t M_{i,t}]^{-2/5} h_T^{-k} \right)$. It then follows along the same lines as in the proof of Proposition 1 that the estimation error becomes negligible. This means that Proposition 1 still holds as long as

$$d_T^{-1} \max \left\{ [\inf_t M_{i,t}]^{-2/5} h_T^{-k}, [\inf_t M_{i,t}]^{-1/5} T^{-1/2} h_T^{-k} \right\}$$

shrinks to zero. In addition, Proposition 2 still holds, replacing M with $[\inf_t M_{i,t}]^{-2/5}$.

⁶ Note that, for notational simplicity, we omit the jittering term correction by implicitly assuming $m = 1$ in Section 2.2 of Barndorff-Nielsen et al. (2011).